

Something just happened. What does it do to our portfolio?

A war, a tariff package, a surprise rate decision, a competitor's refinery explosion, a Category 4 hurricane in the Gulf. A portfolio manager needs three answers in minutes, and needs to trust every number in them.

WHO IS EXPOSED

Which holdings the event reaches: directly, through a competitor, or through a market factor such as oil or rates.

HOW FAR COULD IT GO

What markets did in the five trading days after similar past events, weighted by how similar they were.

WHAT TO DO

Value at risk, the scenario result, and a hedge plan inside position limits, with the risk before and after.

THE DEMO IN ONE SCREEN

TEMPEST

Live

Replay

Hurricane Ida (2021)

As of (UTC)

27/08/2021, 09:00 P

Reliability

Audit

PORTFOLIO

As of Aug 27, 2021, 21:00 UTC

Tempest Multi-Sector Fund

NAV

CASH

\$10M

\$501.9K

ENERGY

XOM

Exxon Mobil

\$680K

- 6%

direct

factor

- sent

+0.33

+1.9%

CVX

Chevron

\$480K

- 4%

direct

factor

- sent

+0.31

+1.5%

XLE

Energy Select Sector SPDR

\$400K

- 4%

direct

factor

- sent

+0.78

+2.7%

REFINER

VLO

Valero Energy

\$480K

- 4%

direct

factor

- sent

+0.29

+2.5%

MPC

Marathon Petroleum

\$380K

- 3%

direct

factor

- sent

+0.29

+3.3%

COMMODITY PROXY

USO

United States Oil Fund

\$280K

- 2%

direct

factor

- sent

+0.78

+1.4%

GOLD

GLD

SPDR Gold Shares

\$499.8K

- 5%

+1.5%

NEM

Newmont

\$280K

- 2%

factor

+2.8%

TECH

MSFT

Microsoft

\$599.7K

- 6%

factor

+0.2%

AAPL

Apple

\$499.9K

- 5%

factor

+0.7%

SEMS

NVDA

NVIDIA

\$499.8K

- 5%

factor

+2.6%

TSM

Taiwan Semiconductor (ADR)

\$299.9K

- 3%

factor

+0.3%

How will the forecasted Category 4 hurricane in the Gulf of Mexico affect our current energy holdings?

Ask

How will the Russian invasion of Ukraine aff...

How will the forecasted Category 4 hurrican...

Our competitor just had a refinery explosion...

What if China blockades Taiwan?

What is moving our portfolio today?

What if it only reaches Category 2?

EVENT

Disaster

Hurricane

Low Severity

Replay

Hurricane Ida

FIRST REPORT

Aug 26, 2021, 18:00 UTC

ARTICLES

50

DOMAINS

37

VOLUME Z / TONE Z

n/a / n/a

News features were unavailable for this event.

DIRECT

MPC

PSX

VLO

XOM

CVX

XLE

USO

PEERS

OXY

SECTORS

energy

refiner

commodity proxy

FACTORS

GULF_GASOLINE

HH_NATGAS

RATES

WTI

MARKET

AGENT GRAPH

pending

running

done

skipped

degraded

failed

planner

event

sentiment

analogs

risk

hedging

synthesizer

verifier

ANSWER

High Confidence

replay

perfect-forecast replay

Hurricane Ida forecast to make landfall Aug 29, 2021 as a Category 4 storm, with 23.1% of Gulf Coast refining capacity at risk and energy holdings poised for modest gains despite conflicting commodity signals.

The hurricane threatens 9 refineries in the impact zone, with Exxon Mobil facing the largest direct exposure at 29.3% of its capacity. The scenario forecasts WTI crude rising +1.5% while Gulf gasoline declines -0.9%, creating headwinds for refiners despite upstream strength; rates are expected to rise, adding pressure to rate-sensitive energy equities. Hedges selling \$1.5M gross notional across SPY, XLF, and CRAK are designed to offset headwinds and stabilize value, providing upside from energy equities, reducing the duration of debt.

Steps

Events

News

Sources

STEP LOG

Succeeded

53 scored articles; 5 sectors, 10 holdings - 5.4 s

claude-haiku-4.5-20251001 - 800 in / 145 out - \$0.0015

+7.6 s Evidence

6 rows (E27 to E32)

+7.6 s Evidence

7 rows (E20 to E26)

+7.6 s Start

analogs started

+11.3 s Done

analogs

28 eligible events, effective sample 18.8; groups type, regime, weather - 3.8 s

claude-haiku-4.5-20251001 - 1,338 in / 228 out - \$0.0025

+11.4 s Evidence

16 rows (E33 to E48)

+11.4 s Start

risk started

+11.4 s Done

risk

22 holdings with an exposure channel; scenario P&L computed - 81 ms

+11.5 s Evidence

60 rows (E49 to E108)

+11.5 s Start

hedging started

+37.4 s Done

hedging

Model plan: 3 actions, scenario result moves from 47242 to 40415 - 25.9 s

claude-haiku-4.5-20251001 - 33,146 in / 1,456 out - \$0.040

+37.5 s Evidence

3 rows (E109 to E111)

+37.5 s Start

synthesizer started

+48.5 s Done

synthesizer

Answer drafted with evidence placeholders - 11.1 s

claude-haiku-4.5-20251001 - 7,989 in / 782 out - \$0.012

+48.6 s Start

verifier started

+48.6 s Done

verifier

All grounding checks passed - 5 ms

+48.6 s Succeeded

Run succeeded - high confidence - 48.6 s - \$0.065

- **Ask in plain English.** Replay mode fixes the clock at an as-of time; nothing later can be read.
- **The graph runs live:** planner, event, then weather, sentiment and macro in parallel, analogs, risk, hedging, answer, verifier.
- **Exposure badges** on every holding: **direct**, **peer**, **factor**.
- **Every number is a chip** linked to its source and as-of time. The verifier counts ungrounded numbers: zero.
- **Hurricane runs** add the storm track and the refineries inside the impact radius.

ARCHITECTURE

Real data in, an audited answer out

AGENT GRAPH (LANGGRAPH, SAME SHAPE ON EVERY RUN)

planner → event → weather · sentiment · macro → analogs → risk → hedging → synthesizer → verifier

DATA

GDELT Alpha Vantage news Tiingo prices FRED

EIA NHC HURDAT2 Open-Meteo

A BullMQ worker ingests on schedules into Postgres; past events and news live in a Pinecone index for similarity search.

TRUST

- Nodes write **evidence rows**; the model writes `{{E12}}` placeholders, never digits.
- The verifier rejects stray digits, unknown symbols, hedges outside limits, missing caveats.
- A failing source degrades one node; the run continues and says what is missing.

DATA FLOW

sources → worker (BullMQ, Redis) → Postgres + Pinecone → api (tRPC, SSE stream) → terminal (Next.js)

TypeScript monorepo: Next.js terminal, Express + tRPC with SSE streaming, Drizzle + Postgres, Redis, Pinecone, Claude.

PROOF, MEASURED BY SCRIPTS (DOCS/RESULTS.MD)

We measured it, including where it is weak

EVAL, 20 QUESTIONS

95%

intent and event type
right; 19 of 20 runs
succeeded, verifier
100% after repair

ROBUSTNESS DRILLS

26/26

runs pass with a source,
the index, a quota or the
model key taken away

RUN TIME

45 s

median run, p95 79 s;
first step done in 4 s

COST PER RUN

\$0.04

Claude tokens, mean of
the eval runs

FORECAST BACKTEST: LEAVE-ONE-OUT OVER 64 PAST EVENTS, 6 TARGETS

Model	Directional accuracy	MAE (5-day)
Same event type	61.4%	5.78%
Market regime (VIX)	56.6%	5.22%
Combined	52.4%	5.23%
Unconditional mean	49.3%	5.49%

Small samples (3 to 31 events per type) and hand-picked events; news features were unavailable on our network, so the news model has no predictions. The reliability page shows every table.

WHAT IS REAL, WHAT IS SIMULATED

REAL

- News from GDELT and Alpha Vantage; daily prices; FRED and EIA series; NHC and HURDAT2 storm tracks; EIA refinery locations.
- A Pinecone index of 64 past events and their market reactions.
- Claude models for planning, scoring news, hedging and writing; every number from computed evidence.
- Replay mode cannot read anything dated after its as-of time.

SIMULATED

- A fixed paper portfolio: 23 positions, \$10M.
- Hedges use ETFs in place of futures; nothing goes to a broker.
- Hurricane replays use the best track as a perfect forecast, and say so.
- Prices are end of day.

From an event to exposure, analogs, risk and a hedge, in under a minute, with every claim traceable.